Free balancing equations worksheet worksheet 14

Maths | 10th grade | Algebra

1. If f(x) = 2x^2 - 3x + 5, find f(2) + f(-1).

A) 8
B) 10
C) 12
D) 16

2. Solve for x in the equation: 3^(2x) = 81.

A) 2
B) 3
C) 4
D) 5

3. The sum of the squares of two consecutive integers is 85. What are the integers?

A) 7 and 8
B) 8 and 9
C) 6 and 7
D) 9 and 10

4. Find the value of k if kx - 2(k+1) = 3x + 4 has no solution.

A) -1
B) 1
C) 2
D) 3

5. Solve the inequality: 5x - 7 < 3x + 1.

A) x < 4
B) x > 4
C) x < -4
D) x > -4

6. If (x + 2)(x - 3) = 0, what is the sum of the solutions?

A) -1
B) 1
C) 5
D) -5

7. The function g(x) is defined by g(x) = x^2 + 4x + c. Find c so that g(x) has a root at x = -2.

A) 0
B) 2
C) 4
D) -4

8. Determine the slope of the line through points (2, 3) and (6, -1).

A) 1
B) -1

C) 2
D) -2

9. Find the product of the roots of the equation x^2 - 5x + 6 = 0.

A) 5
B) 6
C) -5
D) -6

10. If 5a + 2b = 12 and a = 2b, what is the value of b?

A) 2
B) 3
C) 4
D) 5

Answer Key

1. D
Calculate f(2) = 2(2)^2 - 3(2) + 5 = 11 and f(-1) = 2(-1)^2 - 3(-1) + 5 = 5. Thus, f(2) + f(-1) = 11 + 5 = 16.

2. A
Since 81 = 3^4, set 2x = 4, giving x = 2.

3. B
Let integers be x and x+1. x^2 + (x+1)^2 = 85 leads to 2x^2 + 2x = 84. Solving gives x = 8.

4. A
For no solution, coefficients of x must be equal and constants not: k = 3 and -2k - 2 "` 4. Solving gives k = -1.

5. A
Simplify to get 2x < 8, thus x < 4.

6. C
The solutions are x = -2 and x = 3, hence the sum is -2 + 3 = 1.

7. D
Substitute x = -2 into g(x), set to 0: (-2)^2 + 4(-2) + c = 0. Solve to find c = -4.

8. D
Slope m = (-1 - 3)/(6 - 2) = -4/4 = -1.

9. B
For ax^2 + bx + c = 0, the product of roots is c/a = 6/1 = 6.

10. A
Substitute a = 2b into 5a + 2b = 12 to get 10b + 2b = 12, solve for b: 12b = 12, b = 1.